

1. Which method requires the system to have prior information about the resources a process will request and use?

1 point

- ☒ Deadlock Avoidance
- ☐ Deadlock Detection and Recovery
- ☐ Deadlock Prevention
- ☐ Ignoring the problem

2. Which of the following best describes the role of the mapping function $F: R \rightarrow N$ in preventing Circular Wait?

1 point

- ☐ It determines which process should be preempted in case of a deadlock.
- ☐ It tracks the order in which processes request resources.
- ☒ It assigns a unique number to each resource type, enforcing an order in which resources must be requested.
- ☐ It allows processes to request resources in reverse order.

3. How is the Available vector computed in the Banker's Algorithm?

1 point

- ☒ By subtracting the total allocated resources from the total resources
- ☐ By adding the Allocation matrix to the Max matrix
- ☐ By calculating the sum of the resources requested by all processes
- ☐ By adding all elements of the Need matrix

4. What happens in the second step of the Resource Request Algorithm?

1 point

- ☐ The algorithm updates the Allocation matrix with the requested resources.
- ☐ The algorithm rolls back any temporary allocations if the state is not safe.
- ☒ The algorithm checks if the requested resources are currently available.
- ☐ The algorithm checks if the requested resources exceed the process's maximum claim.

5. In the deadlock detection algorithm, what action is taken when a process's request is found to be less than or equal to the Work vector?

1 point

- ☐ Move the process to a waiting state
- ☒ Update Work to include the process's Allocation and set Finish [i] to true
- ☐ Ignore the request and continue
- ☐ Abort the process immediately